Principles of Macroeconomics

Personal Notes

Mihir Rao

Contents

1 Introduction to 14.02 Principles of Macroeconomics

Macroeconomics studies the behavior of an economy as a whole. Its objects are aggregates and the links among them: the United States or China rather than a single household or firm, the inflation rate rather than one good's price, the unemployment rate rather than one worker's employment outcome, and the exchange rate rather than the relative price of two particular goods.

1.1 Why Macro Is Not Just "Lots of Micro"

An economy is made up of households, firms, and individual markets, so it is tempting to treat macroeconomics as the sum of many microeconomic decisions. That construction misses the central difficulty. Once many agents interact, their beliefs, prices, and decisions must fit together in an equilibrium; the aggregate system has relationships that are hard to see by studying an individual component in isolation.

Macroeconomists therefore use deliberately simplified models. The goal is not to reproduce every institutional detail, but to retain the mechanism that matters for the question at hand. This is a judgment as well as a technical exercise: a model should be simple enough to reveal the essential aggregate relationship without becoming so simple that it explains the wrong thing.

> ***Note:*** *The course uses modest mathematics. A concept may be subtle, but the aim is to understand the economic forces, not to turn the introductory course into a PhD-level modelling sequence.*

1.2 What the Course Is Meant to Make Legible

The practical objective is economic literacy. By the end of the course, a student should be able to read an IMF *World Economic Outlook* or a financial newspaper, identify the macroeconomic variables at issue, and evaluate the proposed explanation rather than merely accept it. The course is therefore organized around recurring current events. As the vocabulary and models develop, the same news story can be revisited with a more precise account of its causes and consequences.

Formal definitions come later. This opening lecture instead introduces the kind of evidence macroeconomists use and the questions they ask of it:

- Why do inflation and wage growth move together, and when should that connection concern policymakers?
- Why does unemployment rise in recessions and recover at different speeds after different shocks?
- How can a central bank influence spending, investment, and asset values through the interest rate?

- Why can the same worldwide episode have a common core but different country-specific amplifiers?

1.3 A Macroeconomic Snapshot: Inflation after COVID-19

The lecture uses the early-2023 U.S. economy as a motivating case. The unemployment rate rises during recessions; the Great Recession was followed by a long labor-market recovery, while the sharp COVID-19 disruption was followed by a much faster rebound. By the period discussed in the lecture, unemployment was exceptionally low and wage growth was rapid, especially in accommodation and food services.

Those facts are welcome from an individual worker's perspective, but macro analysis asks about their joint implications. Wage growth and inflation tend to be strongly related. The usual inflation objective for large advanced economies is roughly 2% per year, whereas the lecture describes readings around 6.5% to 8% in the United States. Inflation had begun to ease, but remained far above that benchmark.

Definition: *Inflation* is the rate at which the overall price level rises. It is an aggregate concept: it is not the price increase of one product. A precise measurement definition is deferred to the next lecture.

The post-COVID inflation episode was broad, not uniquely American. Many economies that had previously experienced inflation near or below 2% saw it rise substantially. The shared backdrop was an unusually strong recovery; local conditions then added their own effects. In Europe, for example, the war in Ukraine and the associated energy-price shock intensified inflation.

> ***Note:*** *Numbers in this section are the lecture's contemporaneous examples, not current estimates. Their role is to show how macroeconomic data become a policy problem.*

1.4 Interest Rates, Aggregate Demand, and Financial Markets

The central bank's principal instrument for controlling inflation is the interest rate. Lower interest rates make borrowing less expensive, which tends to support consumption and firms' investment. Higher rates work in the opposite direction, restraining demand. This trade-off explains the rapid rate increases undertaken once policymakers judged inflation to be too high.

Monetary policy also reaches financial markets. In the lecture's example, the stock market fell sharply when COVID-19 first appeared, then rose well above its pre-pandemic level while monetary policy was very loose. Later rate hikes were associated with a broad decline in equity values. The point is not that one chart proves a complete causal story, but that interest rates change the discounting and financing conditions relevant to many assets.

Financial prices can move immediately when new information arrives because they embody expectations about the future. An employment release, for example, can change investors' forecasts of inflation and future monetary policy before the stock market has even opened. Macroeconomics supplies the framework for tracing those links rather than treating each data release as an isolated headline.

1.5 Takeaways

- Macro studies aggregate outcomes and the interactions that connect them; it needs models tailored to those interactions.
- Good macro models make disciplined simplifications so that the key mechanism is visible.
- Inflation, unemployment, wage growth, interest rates, and asset prices are jointly informative, not separate scorecards.
- Current events are both an application of macroeconomic tools and a source of questions that motivate building those tools.

Source: MIT 14.02 Principles of Macroeconomics, Spring 2023, Lecture 1 video by Ricardo J. Caballero.

2 Basic Macroeconomic Concepts

Macroeconomics needs measures that compress a vast collection of different goods and services into a small number of interpretable aggregates. That is harder than measuring the output or price of one firm: the economy produces cars, medical care, financial services, restaurant meals, and many other things at once. This lecture introduces the accounting conventions that make aggregate output, labor-market conditions, and inflation measurable.

2.1 Aggregate Output and GDP

The modern national accounts provide a systematic measure of aggregate economic activity. Their central output measure is *gross domestic product* (GDP). In these notes, "goods" is shorthand for goods *and* services: services are an important part of what households consume even when they are not tangible objects.

Definition 2.1 (Gross domestic product). GDP is the market value of final goods and services produced within an economy during a specified period. It is a *flow*: a statement such as "GDP in 2022" refers to production over that year, not to a stock measured at one instant.

The word *final* prevents double counting. Suppose a steel producer sells steel worth \$100 to a car producer, which uses it to sell cars worth \$200 to households. Adding the two firms' sales would report \$300, even though the steel is already embodied in the cars. The final-goods method therefore records GDP as \$200. This answer should not depend on whether the two firms are separately owned or are merged into one vertically integrated company.

There are three equivalent accounting perspectives:

1. **Final-expenditure method.** Sum the market value of final goods and services produced during the period.
2. **Value-added method.** For each producer, subtract purchased intermediate inputs from sales, then add the resulting value added across producers. In the steel–car example, the steel firm adds \$100 and the car firm adds \$100, again yielding \$200.
3. **Income method.** Sum the income generated by production: compensation to labor, returns to capital, and, in a fuller account, taxes and other claims. What firms produce becomes somebody's income.

> ***Note:*** *An intermediate input is a good or service purchased to make another good or service. A firm's own workers and its use of its own capital are factors of production, not purchased intermediate goods. Contracting an outside firm does not evade the accounting rule: that firm's value added is counted at its own stage of production.*

The equality between aggregate production and aggregate income is especially useful in macroeconomics. In a closed-economy abstraction, income generated by producing the aggregate good is ultimately spending power for that same aggregate output. Later models use this circular connection to study how output is determined.

2.2 Nominal and Real GDP

For final goods indexed by i, nominal GDP in period t is

$$Y_t^{\text{nom}} = \sum_i p_{i,t} q_{i,t},$$

where $p_{i,t}$ is the current price and $q_{i,t}$ is the quantity produced. It is measured in current dollars, so it can rise either because the economy produces more or because prices are higher. Those are economically distinct changes.

Real GDP holds prices fixed at a chosen base period b:

$$Y_t = \sum_i p_{i,b} q_{i,t}.$$

It therefore tracks changes in quantities at common prices. Nominal and real GDP coincide in the base period because both use that period's prices. In ordinary discussion, "GDP" usually means real GDP when the object of interest is the economy's physical output and growth.

Example: Consider an economy that produces only cars. It produces 10, 12, and 13 cars in 2011–2013, while the car price is $20,000, $24,000, and $26,000. Its nominal GDP is respectively $200,000, $288,000, and $338,000. Taking 2012 as the base year, real GDP is $240,000, $288,000, and $312,000. The gap between nominal and real growth reflects rising prices; the real series keeps the 2012 price of $24,000 throughout.

Fixed-base calculations are a useful conceptual device. In practice, national accountants also confront quality change: a more capable computer may cost more because it is a better product, not simply because the general price level has risen. Such measurement choices matter when converting dollar values into a measure of real output.

2.3 Output Growth and Recessions

Real GDP growth describes how aggregate production changes from one period to the next. A decline in real output is a key sign of a recession, although a recession is a broader judgment about a significant and widespread downturn; the popular "two consecutive quarters of negative real GDP growth" rule is a convenient heuristic rather than a complete definition. The sharp 2020 contraction and subsequent reopening illustrate why growth rates should be read alongside the event that generated them.

Over longer horizons, nominal growth can be a deeply misleading indicator. Persistent inflation makes a dollar-valued series rise even if real activity stagnates or contracts. The distinction is consequently most important when comparing periods with substantially different price levels.

2.4 Employment, Unemployment, and Participation

Employment counts people with jobs. Unemployment is narrower than non-employment: a person is counted as unemployed only if they do not have a job *and* are actively looking for one. Let E denote employment and U unemployment. The labor force is

$$L = E + U,$$

and the unemployment rate is

$$u = \frac{U}{L}.$$

The denominator is the labor force, not the entire population. People without jobs who are not searching are classified as outside the labor force, so they do not enter either U or L.

In the United States, these categories are estimated with the Current Population Survey of households. Survey classifications are essential to the definition: someone who has stopped searching after a long unsuccessful spell may not be counted as unemployed even though their economic circumstances remain relevant. For this reason, analysts also inspect broader measures of non-employment.

The labor-force participation rate complements unemployment:

$$\text{participation rate} = \frac{L}{\text{working-age population}}.$$

It asks how many people who could in principle work are participating in the labor market. A falling participation rate can coexist with a low unemployment rate, because a smaller number of people are seeking work. The lecture uses the post-COVID recovery as a dated example: strong demand and a slower-than-expected return to the labor force both made the labor market tight. Those contemporaneous observations motivate, but do not by themselves establish, later links among labor markets, output, and inflation.

2.5 Price Levels and Inflation

Definition 2.2 (Inflation). Inflation is a sustained increase in the general price level. It is not a one-off increase in the relative price of one good. If P_t is a chosen price index, its period-to-period inflation rate is

$$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}}.$$

Deflation is negative inflation: a sustained fall in the general price level.

There is no single price index for every purpose. The GDP deflator is a broad economy-wide index implied by the national accounts:

$$P_t^{\mathrm{GDP}} = \frac{Y_t^{\mathrm{nom}}}{Y_t}.$$

It compares nominal GDP with real GDP. The consumer price index (CPI), by contrast, follows the prices relevant to a consumer basket and is the measure most commonly reported in public discussion. The two can differ because they cover and weight goods differently, but both distinguish changes in prices from changes in real quantities.

Measures can also be tailored to a question. Core CPI, for example, excludes food and energy because those components are often volatile. This does not make the excluded prices irrelevant; it is a way to reduce short-run noise when assessing persistent price pressure.

2.6 Why These Definitions Matter

The lecture closes by contrasting long-run macroeconomic paths. These measures supply the common language for later models of output, policy, labor markets, and inflation.

Source: MIT OCW 14.02, Spring 2023, Lecture 2.

3 The Goods Market

The first model of the course explains short-run output through demand in the goods market. The emphasis is deliberately narrow: prices, interest rates, and productive capacity are held in the background so that the feedback between spending, output, and income is visible.

3.1 The Short-Run Demand Assumption

In the very short run, firms can often vary production without immediately changing their productive capacity. The model therefore treats output as demand-determined: when planned spending on domestically produced goods changes, firms adjust production to meet it. Since production creates income, that output adjustment changes income and thereby feeds back into spending.

This is not a claim about every horizon. Over longer periods, capital, labor, technology, and prices constrain what an economy can produce. The present model instead isolates the short-run mechanism that makes a fall in demand a source of lower output and a possible recession.

3.2 Aggregate Demand

For a general open economy, demand for domestic production is

$$Z = C + I + G + X - IM,$$

where C is consumption, I investment, G government purchases, X exports, and IM imports. Imports are subtracted because they are spending by domestic households, firms, or governments on foreign rather than domestic production.

The initial closed-economy version of the model sets $X = IM = 0$, giving

$$Z = C + I + G.$$

Government purchases are purchases of goods and services. A transfer payment, such as a check sent to a household, is not itself G: it affects demand only when the recipient changes consumption. The model takes I, G, and taxes T as given parameters; consumption supplies the behavioral link that makes aggregate demand respond to income.

3.3 Consumption and Disposable Income

Disposable income is income after taxes:

$$Y_D = Y - T,$$

where Y denotes real output and, by national-income accounting, aggregate income. The simple consumption function is

$$C = C_0 + C_1(Y - T), \qquad 0 < C_1 < 1.$$

Definition 3.1 (Autonomous consumption). C_0 is the component of consumption that is independent of current disposable income in this model. It summarizes omitted influences such as household confidence, wealth, and expectations.

Definition 3.2 (Marginal propensity to consume). C_1 is the fraction of one additional unit of disposable income that is spent on consumption. If $C_1 = 0.6$, an additional dollar of disposable income raises consumption by \$0.60 and saving by \$0.40.

The linear specification is an approximation, not a complete theory of household behavior. Its value is that it makes the feedback transparent: higher output raises income, part of the additional income is spent, and that extra spending raises demand for output.

3.4 Goods-Market Equilibrium

Substituting the consumption function into aggregate demand yields

$$Z = C_0 + C_1(Y - T) + I + G.$$

This equation is an aggregate-demand *function*; it describes desired spending at every possible level of output. Equilibrium adds a distinct condition:

$$Y = Z.$$

At equilibrium, production equals planned demand. Away from equilibrium, the two need not be equal: unintended inventory changes give firms a reason to alter production.

Solving the two equations gives equilibrium output:

$$Y = \frac{1}{1 - C_1}\big(C_0 + I + G - C_1 T\big).$$

The term in parentheses collects the autonomous components of spending, including the tax effect on consumption. The model thus treats a change in confidence, investment, purchases, or taxes as a shift in aggregate demand, not as an immediate change in productive capacity.

Note: *The standard diagram places Y on the horizontal axis and Z on the vertical axis. The 45-degree line represents $Z = Y$. The aggregate-demand line has slope C_1, so it is flatter than the 45-degree line when households save a positive share of an additional dollar. Their intersection is the unique goods-market equilibrium in this simple model.*

3.5 The Multiplier

The coefficient

$$k = \frac{1}{1 - C_1}$$

is the *multiplier*. Because $0 < C_1 < 1$, it exceeds one. A one-unit increase in autonomous spending raises output by more than one unit:

$$\Delta Y = k\,\Delta A,$$

where ΔA is an autonomous shift such as a change in C_0, I, or G.

The amplification comes from repeated rounds of expenditure. For example, an initial $1 billion increase in government purchases first raises output and income by $1 billion. If $C_1 = 0.5$, households spend an additional $0.5 billion; that becomes another firm's income and output, which generates a further $0.25 billion of consumption, and so on. The geometric sequence sums to $2 billion, precisely the multiplier $1/(1 - 0.5)$ times the initial increase.

A higher marginal propensity to consume produces a larger multiplier because more of every income increase recirculates as spending. If households save all of an additional dollar, the feedback stops after the initial round and the multiplier is one.

Taxes affect demand indirectly through disposable income. Holding other parameters fixed,

$$\Delta Y = -\frac{C_1}{1 - C_1}\Delta T.$$

An increase in taxes reduces consumption at each level of Y and shifts the aggregate-demand line down. Its absolute effect is smaller than an equal change in government purchases in this model because households spend only the fraction C_1 of a tax change.

3.6 Saving and the IS Interpretation

The same equilibrium can be expressed using saving. Private saving is disposable income not consumed,

$$S_{\text{private}} = Y - T - C,$$

and government saving is

$$S_{\text{government}} = T - G.$$

Adding the two gives total saving. Rearranging the goods-market equilibrium $Y = C + I + G$ shows that, in this closed-economy model,

$$I = S_{\text{private}} + S_{\text{government}}.$$

This is an equivalent accounting route to equilibrium, not a separate model. It foreshadows the later IS relation: the goods market is in equilibrium when investment and saving are equal.

3.7 The Paradox of Thrift

Individual saving can be prudent, but a simultaneous attempt by all households to save more can be contractionary in the short run. In this model, a stronger desire to save at every income level is a fall in C_0. Aggregate demand falls, firms reduce output, income declines, and the lower income offsets the initial attempted increase in saving until $I = S$ is restored.

The result is a *paradox of thrift*: with investment and government saving fixed, a collective reduction in consumption can lower output rather than raise equilibrium saving. It is a short-run aggregate result, not advice that an individual should never save.

3.8 Fiscal Policy

Fiscal policy operates through government purchases and taxes. An increase in G or a decrease in T is expansionary because it shifts aggregate demand up; the multiplier then amplifies the initial impulse. A decrease in G or an increase in T is contractionary because it lowers demand and equilibrium output. Which direction is desirable depends on the economy's condition: the same demand-reducing policy that aggravates a recession can help cool an overheated economy.

Source: MIT OCW 14.02, Spring 2023, Lecture 3.